

1) Backcasting:

Backcasting involves using a time series model to forecast past values, typically for periods where data is missing or unreliable.

• Purpose:

Backcasting serves several purposes:

- 1- Data imputation: Replacing missing values with estimated ones.
- 2- Data validation: verifying the accuracy of existing data.
- 3- Historical analysis: Analyzing past trend and patterns.
- 4- Model testing: Evaluating model performance on historical data.

• Common backcasting methods include:

- 1- ARIMA model (Autoregressive integrated moving average).
- 2- Exponential smoothing (ES).

2) overfitting:

overfitting in t.s models occurs when a model is too complex and captures noise instead of underlying patterns, resulting in poor forecasting performance.

→ Effects of overfitting in T-S model:

- 1- Poor forecasting accuracy.
- 2- Model is difficult to interpret.

→ Causes:

- 1- Insufficient training data.
- 2- Noise in the data.
- 3- Inadequate model selection.

3. Postmantau test

The Postmantau test is a statistical test used in T.S analysis to:
check for autocorrelation. • validate model adequacy.
detect non-randomness.

Hypotesis:

$$H_0: \rho_1 = \rho_2 = \dots = \rho_m = 0$$

Types of Postmantau test:

i- Box-Pierce test: (1970)

The box-pierce statistics is a widely used diagnostic tool in t.s analysis to test for:

1- Autocorrelation. (2) Partial autocorrelation. (3) non-randomness.

Statistics is: $Q_m = n \sum_{k=1}^m \hat{\rho}_k^2 \sim \chi^2(df)$

where " $\hat{\rho}_k$ " is the residuals autocorrelation at lag k.

ii- Liung-Box Test (1972)

$$Q_m = n(n+2) \sum_{k=1}^m \frac{\hat{\rho}_k^2}{n-k} \sim \chi^2(df)$$

Liung-Box test is used to test autocorrelation at multiple lag (same Box-pierce test).

uses:

- P. test is used in t.s to identify autocorrelation.
- P. test is used in t.s to detect non-random pattern.
- P. test is used in t.s for diagnostic checking, i.e. check for stationarity • verify normality of residual.
- P. test is used in t.s for detecting seasonal patterns.
- P. test is used in t.s to evaluate forecasting accuracy.

4- Extrapolation of trend curve in T.S. :-

Extrapolation of a trend curve in T.S. involves extending the curve beyond the available data to forecast future values.

Types :

- 1- Linear extrapolation.
- 2- Non-linear extrapolation.
- 3- Polynomial extrapolation.

Methods :

- Regression analysis: Use linear or non-linear regression model.
- Exponential smoothing: weighted moving averages.
- ARIMA/SARIMA models: Autoregressive integrated M.A.

5 Forecasting :-

Forecasting is the process of predicting future values of a T.S. based on past and present data. Forecast is objective as well as subjective approach. Forecasting involves risks due to forecast errors and these forecast errors depends on the forecast lead time.

Types :

- 1- Qualitative forecast: Uses expert judgement & opinions.
- 2- Quantitative forecast: Uses statistical models and data.

Forecasting procedure:

Naive methods:

t	1	2	3	4	5	6	7	8	9	10	11
y_t	y_1	y_2	y_3	y_4	y_5	y_6	y_7	y_8	y_9	y_{10}	?

- 1- Most Recent Observation:

$$\hat{y}_{11} = y_{10} = 1 \cdot y_{10} + 0 \cdot y_9 + \dots + 0 \cdot y_1$$

2- Mean Method:

$$\hat{y}_n = \bar{y} = \frac{1}{10} (y_1 + y_2 + \dots + y_{10})$$

3- Mean of 'm' most recent observations:

$$\hat{y}_n = \frac{1}{3} (y_{10} + y_9 + y_8) \quad \text{where } m=3$$

First method gives '1' weight to most recent values and '0' weight to remaining.

2nd method gives equal weights to all the values.

3rd method gives 'm' weights to some most recent values and '0' weights to remaining.

4- Exponential Smoothing:

Simple exponential smoothing is a popular forecasting technique in t.s. models. SES is not a subjective approach. It is a statistical procedure but choice α ourself. when we go only last 5-years then select α close to 1 but when we go last 15 years to 20 years then select α close to zero.

$$\text{Suppose, } X_n(t) = C_0 X_n + C_1 X_{n-1} + C_2 X_{n-2} + \dots + C_{n-1} X_1$$

C 's are weights.

b- Forecast ~~role~~ error and its role in updating the forecast:

Forecast error is the difference b/w the actual or real and the predicted or forecast value of a time series or any other phenomena of interest.

Types of forecast error:

1- Mean absolute error (MAE).

2- Mean squared error (MSE).

3- Root mean squared error (RMSE).

Role

1- Error correction: update forecasts by adjusting

for past errors.

2- Model selection: Choose the best model based on error metrics.

3- Forecast Combination: Combine multiple models to reduce error.

7- Updating forecast:

Updating forecasts involves revising prediction based on: 1) new data. 2) changes in underlying pattern. 3) model performance evaluation.

Types:

- Rolling forecasts: Regularly update forecasts with new data.
- Model selection: Switch to a better performing model.

Importance:

- 1- Improves accuracy: Incorporates new information, reducing forecast error.
- 2- Adapts to changes: Adjusts to shifts in trends, seasonality or outliers.
- 3- Optimize resource allocation: Allocates resources efficiently based on updated forecasts.

8- Model selection criterion:

- 1- Akaike information criterion (AIC)
- 2- Bayesian information criterion (BIC)
- 3- Schwarz Bayesian information criteria (SBIC)

AIC:

1/ AIC is the method of model selection criterion in this analysis.

2/ To apply this approach a range of potential

ARIMA models are estimated by maximum likelihood methods.

For each model the criteria of AIC is given by-

$$AIC_{\text{par}} = -2 \ln(\text{mx. likelihood}) + 2k \approx \ln(\hat{\sigma}_z^2) + k \frac{2}{n} + \text{Constant}$$

Here $\hat{\sigma}_z^2$ is maximum likelihood estimate of σ_z^2

and $k = p + q + 1$ is a number of estimated parameters, including a constant term.

AIC has a weaker penalty for complexity.

BIC

Point 1, 2, 4 are same as AIC.

$$BIC_{\text{par}} = \ln(\hat{\sigma}_z^2) + k \frac{\ln(n)}{n}$$

BIC has a strong penalty for complexity.

9-Box-Jenkins model building approach:

In time series analysis the most imp. steps are to identify and build a model based on the available data.

The box-Jenkins approach is a systematic methodology for building and evaluating time series models, developed by George Box and Guilym Jenkins in 1976.

Stages/steps for model building:

1- Observed time series

The first step is to plot the data. Through this we usually get a good idea about whether the series contains a trend, seasonality, outliers and non-constant variances etc.

2- Postulate General ARMA model; we plot autocorrelation and partial autocorrelation function

which are the identification tools for the model.

3- Identify the possible model.

- $ARMA(P, q)$.

- $ARMA(P, 0) = AR(P)$.

- $ARMA(0, q) = MA(q)$.

4- Estimation of model: / methods for estimation of ^{model} AR-

- MLE ii) LSE iii) M.E

5- model Diagnostic checking:

- Residual analysis:

- Diagnostic tests.

(10)

- If model diagnostic checking fails then go to identification.

- If model diagnostic checking passes then take it as final model which is used for forecasting.

10) Model Diagnostic checking:

- This is the last stage of box-Jenkins model building approach. In this stage we critically evaluate the fitted model.

- There are several approach for diagnostic checking. Some methods are given.

1. Residual analysis/plots:

- Residual analysis is a statistical techniques that is used to detect the error in t.s model.

- Residual analysis in t.s model building involves examining the residuals (the difference b/w observed and predicted values) to check if they behave like white noise. This means the

Residual should be uncorrelated, have a constant mean and variance and follow a normal distn.

We plot the residuals and check if there is any pattern still present in it.

2- Diagnostic test:

postmantel test.

Box-Pierce test [Already explained].

II- Model performance measure.

R^2 • MSE • MAPE.

$-R^2$

Measures: Goodness of fit, proportion of variance explained by the model.

Range: 0 (poor fit) to 1 (perfect fit).

Interpretation: High R^2 (70-7): model explains most of the variance.

Low R^2 (<0.3): model explains little of the variance.

MSE: Mean squared error:

Measures: Average squared difference b/w predicted & actual values.

Range: 0 (perfect fit) to ∞ (poor fit).

Interpretation: Low MSE: model predictions are close to actual values.

High MSE: model predictions are far from actual values.

MAPE: Mean absolute %age error:

Measures: Average %age difference b/w predicted and actual values.

Range: 0 (perfect fit) to ∞ (poor fit).

Interpretation: Low MAPE: model prediction are close to actual values.

High MAPE: model prediction are far from actual values.

Given the following information estimate the parameters of AR(2) model process. $\bar{y} = 0.21$, $\gamma_1 = 0.87$, $\gamma_2 = 0.83$.

Sol:

$$\hat{\phi}_1 = \frac{\gamma_1 - \gamma_1 \gamma_2}{1 - \gamma_1^2} = \frac{\gamma_1(1 - \gamma_2)}{1 - \gamma_1^2} = \frac{(0.87)(1 - 0.83)}{1 - (0.87)^2} = 0.6084$$

$$\hat{\phi}_2 = \frac{\gamma_2 - \gamma_1^2}{1 - \gamma_1^2} = \frac{0.83 - (0.87)^2}{1 - (0.87)^2}$$

$$\hat{\phi}_2 = 0.30070$$

$$\hat{\mu} = \frac{\bar{y} - \hat{\phi}_1 \bar{y} - \hat{\phi}_2 \bar{y}}{1 - \hat{\phi}_1 - \hat{\phi}_2}$$

$$= \frac{0.21 - (0.6084)(0.21) - (0.30070)(0.21)}{1 - 0.6084 - 0.30070}$$

$$\hat{\mu} = 0.21$$

Yule-Walker Estimation of AR process.

Consider an AR(p) process:

$$y_t = \phi_1 y_{t-1} + \phi_2 y_{t-2} + \dots + \phi_p y_{t-p} + z_t$$

We know that for an AR(p) process

$$y_k = \phi_1 y_{k-1} + \phi_2 y_{k-2} + \dots + \phi_p y_{k-p}$$

Using theory of method of moment estimation,

$$\gamma_k = \hat{\phi}_1 \gamma_{k-1} + \hat{\phi}_2 \gamma_{k-2} + \dots + \hat{\phi}_p \gamma_{k-p}$$

$$\text{put } k = 1, 2, \dots, p. \quad \gamma_0 = 1, \quad \gamma_{-k} = \gamma_k.$$

$$\gamma_1 = \hat{\phi}_1 + \hat{\phi}_2 \gamma_1 + \dots + \hat{\phi}_p \gamma_{p-1}$$

$$\gamma_2 = \hat{\phi}_2 \gamma_1 + \hat{\phi}_2 + \dots + \hat{\phi}_p \gamma_{p-2}$$

⋮

$$\gamma_p = \hat{\phi}_1 \gamma_{p-1} + \hat{\phi}_2 \gamma_{p-2} + \dots + \hat{\phi}_p$$

$$\begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_p \end{bmatrix} = \begin{bmatrix} 1 & \gamma_1 & \dots & \gamma_{p-1} \\ \gamma_1 & 1 & \dots & \gamma_{p-2} \\ \vdots & \vdots & \ddots & \vdots \\ \gamma_{p-1} & \gamma_{p-2} & \dots & 1 \end{bmatrix} \begin{bmatrix} \hat{\phi}_1 \\ \hat{\phi}_2 \\ \vdots \\ \hat{\phi}_p \end{bmatrix}$$

$$\underline{x} = R \cdot \hat{\underline{\phi}}$$

$$\hat{\underline{\phi}} = R^{-1} \underline{x}$$

AR(1) PROCESS:

$$s_k = \phi_1 s_{k-1}$$

$$x_k = \hat{\phi}_1 x_{k-1}$$

put $k=1$

$$x_1 = \hat{\phi}_1$$

AR(2)

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$$s_k = \phi_1 s_{k-1} + \phi_2 s_{k-2}$$

$$x_k = \hat{\phi}_1 x_{k-1} + \hat{\phi}_2 x_{k-2}$$

$$\text{for } k=1 : x_1 = \hat{\phi}_1 + \hat{\phi}_2 x_1$$

$$\text{for } k=2 : x_2 = \hat{\phi}_1 x_1 + \hat{\phi}_2$$

$$\begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 1 & \gamma_1 \\ \gamma_1 & 1 \end{bmatrix} \begin{bmatrix} \hat{\phi}_1 \\ \hat{\phi}_2 \end{bmatrix} \Rightarrow \begin{bmatrix} \hat{\phi}_1 \\ \hat{\phi}_2 \end{bmatrix} = \begin{bmatrix} 1 & \gamma_1 \\ \gamma_1 & 1 \end{bmatrix}^{-1} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$$

$$A^{-1} = \frac{\text{Adj}(A)}{\det(A)}$$

$$\det(A)$$

$$\begin{bmatrix} \hat{\phi}_1 \\ \hat{\phi}_2 \end{bmatrix} = \frac{1}{1-\gamma_1^2} \begin{bmatrix} 1 & -\gamma_1 \\ -\gamma_1 & 1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \frac{1}{1-\gamma_1^2} \begin{bmatrix} x_1 - \gamma_1 x_2 \\ -\gamma_1 x_1 + x_2 \end{bmatrix}$$

$$\hat{\phi}_1 = \frac{x_1 - \gamma_1 x_2}{1-\gamma_1^2} \quad \& \quad \hat{\phi}_2 = \frac{-\gamma_1 x_1 + x_2}{1-\gamma_1^2}$$

$$\hat{\phi}_1 = \frac{x_1(1-\gamma_2)}{1-\gamma_1^2} \quad \& \quad \hat{\phi}_2 = \frac{x_2 - \gamma_1^2}{1-\gamma_1^2}$$

MA(1)

Describe procedure for finding iterative LSE

Sol: $y_t = \mu + z_t + \theta_1 z_{t-1}$

$z_t = y_t - \mu - \theta_1 z_{t-1}$

$S = \sum_{t=1}^n z_t^2 = \sum_{t=1}^n (y_t - \mu - \theta_1 z_{t-1})^2$

we can't perform analytical optimization due to presence of z_{t-1} in obj-function. one can perform numerical optimization by following algorithm.

Iteration - 0

$z_1^{(0)} = y_1 - \mu^{(0)} - \theta_1^{(0)} z_0^{(0)}$

$z_2^{(0)} = y_2 - \mu^{(0)} - \theta_1^{(0)} z_1^{(0)}$

$\vdots$

$z_n^{(0)} = y_n - \mu^{(0)} - \theta_1^{(0)} z_{n-1}^{(0)}$

Calculate sum of squared errors:

$S^{(0)} = \sum_{t=1}^n (y_t - \mu^{(0)} - \theta_1^{(0)} z_{t-1}^{(0)})^2 = \sum_{t=1}^n (z_t^{(0)})^2$

Iteration - 1

Update the $(\mu^{(0)}, \theta_1^{(0)})$.

$z_1^{(1)} = y_1 - \mu^{(1)} - \theta_1^{(1)} z_0^{(1)}$

$z_2^{(1)} = y_2 - \mu^{(1)} - \theta_1^{(1)} z_1^{(1)}$

$z_n^{(1)} = y_n - \mu^{(1)} - \theta_1^{(1)} z_{n-1}^{(1)}$

Thus we calculate:

$S^{(1)} = \sum_{t=1}^n (z_t^{(1)})^2$

- This process continues till S converges to minimum.
- The value of μ & θ at the minimum level of S are taken as LSEs.

Following the autocorrelation obtain for the residual by fitting an ARMA(2,1) to an observe L.S with $n = 200$.

K	1	2	3	4	5
δ_k	0.04	-0.13	0.25	-0.01	0.09

state hypothesis. Test goodness of fit using
Portmanteau test for $m=5$, $\alpha=0.05$

Sol:

1- Hypothesis:

$$H_0: \beta_k = 0 \quad \text{vs} \quad H_1: \beta_k \neq 0 \quad \text{for } k=1, 2, \dots, n.$$

2- L.O.F : $\alpha=0.05$

3- Test statistics:

$$Q_m = n \sum_{k=1}^m \hat{\epsilon}_k^2 \quad \text{and} \quad Q_m^* = n(n+2) \sum_{k=1}^m (\hat{\epsilon}_k^2 / (n-k))$$

4- Calculation:

$$m=5, \quad n=200$$

$$Q_5 = 200 [0.0892]$$

$$Q_5 = 17.84$$

$$Q_5^* = 200(200+2) [0.0892/195]$$

$$= 40400(0.000457)$$

$$Q_5^* = 18.43$$

5- Critical region:

$$\chi^2 \geq \chi^2_{\alpha, m-p-q} \Rightarrow \chi^2 \geq \chi^2_{0.05(5-2-1)}$$

$$\chi^2 \geq \chi^2_{0.05(3)} \Rightarrow \chi^2 \geq 5.99.$$

6- Conclusion:

Since the computed value $Q_5 = 17.84$ & $Q_5^* = 18.43$ falls in critical region, so we can reject H_0 .

Also, we can concluded that ARMA(2,1) Can't apply in this situation